

Security Audit Report

OWASP MASVS v2 Compliance Assessment



65.0 / 100

Application: UnCrackable-Level1.apk

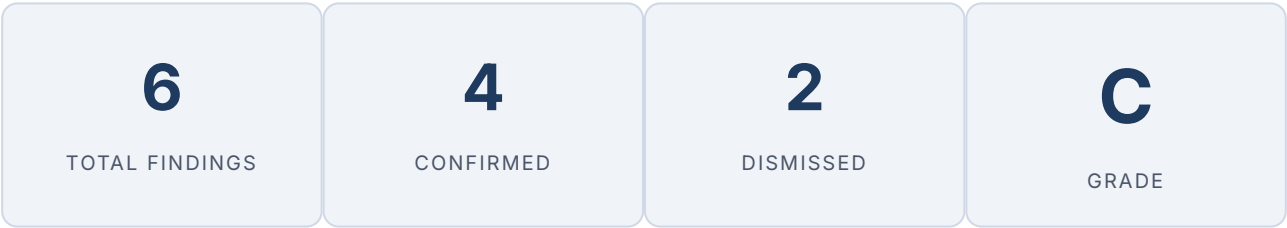
Date: May 01, 2026

Standard: OWASP MASVS v2 / MASTG

Tool: MASVS Audit Copilot

Executive Summary

The security assessment of UnCrackable-Level1.apk resulted in a score of 65.0/100 (Grade C). Out of 6 total findings, 4 were confirmed as real vulnerabilities and 2 were dismissed as false positives by AI triage. The primary risk identified is: App can be installed on a vulnerable unpatched Android version 4.4-4.4.4, [minSdk=19].
Recommended action: Raise minSdkVersion to API 29+ and audit WebView configurations.



SEVERITY	COUNT
Critical	0
High	3
Medium	1
Low	0
Informational	0

Security Posture Dashboard

Most issues are concentrated in Data Storage, Cryptography and Platform Interaction.

MASVS-STORAGE Data Storage ISSUES	MASVS-CRYPTO Cryptography ISSUES	MASVS-AUTH Authentication & Authorization NOT TESTED
MASVS-NETWORK Network Communication NOT TESTED	MASVS-PLATFORM Platform Interaction ISSUES	MASVS-CODE Code Quality NOT TESTED
MASVS-RESILIENCE Anti-Tampering & Reverse Engineering PASS		

Priority Action Plan

Fix This Week

PRIORITY	ISSUE	SEVERITY	MASVS	EFFORT
P1	App can be installed on a vulnerable unpatched Android version 4.4-4.4.4, [mi...	HIGH	MASVS-PLATFORM-3	1-2 days
P1	Calling Cipher.getInstance("AES") will return AES ECB mode by default. ECB mo...	HIGH	MASVS-CRYPTO-1	1-2 days
P1	Application Data can be Backed up [android:allowBackup=true]	HIGH	MASVS-STORAGE-1	1-2 days

Fix This Month

PRIORITY	ISSUE	SEVERITY	MASVS	EFFORT
P1	The App logs information. Sensitive information should never be logged.	MEDIUM	MASVS-STORAGE-1	3-5 days

Attack Scenarios

The following scenarios illustrate how confirmed vulnerabilities could be chained into real-world attacks.

Local Data Extraction via ADB

Attacker access: Physical / ADB access | 2 related findings

An attacker with USB debugging access can extract application data via ADB backup (Application Data can be Backed up [android:allowBackup=true] + The App logs information. Sensitive information should never). This exposes locally stored sensitive information without needing to root the device.

Cryptographic Data Exposure

Attacker access: Local / Reverse engineering | 1 related finding

Weak cryptographic implementations (Calling Cipher.getInstance("AES") will return AES ECB mode b) allow an attacker who reverse-engineers the APK to decrypt locally stored sensitive data or forge cryptographic signatures.

Platform Exploitation via Outdated SDK

Attacker access: Remote / Local | 2 related findings

The application targets outdated Android versions (App can be installed on a vulnerable unpatched Android versi + Application Data can be Backed up [android:allowBackup=true]), exposing it to known platform vulnerabilities. Combined with WebView misconfigurations, this may allow remote code execution.

Technical Findings

4 confirmed vulnerabilities. Dismissed findings are in the Appendix.

Calling Cipher.getInstance("AES") will return AES ECB mode by default. ECB mode is known to be weak as it results in the same ciphertext for identical blocks of plaintext.

HIGH

MASVS: MASVS-CRYPTO-1**CVSS:** 6.5**File:** sg/vantagepoint/a/a.java **AI CONFIRMED**

Description

Calling Cipher.getInstance("AES") will return AES ECB mode by default. ECB mode is known to be weak as it results in the same ciphertext for identical blocks of plaintext.

Business Impact

Weak cryptographic implementation may allow attackers to decrypt sensitive data or forge signatures.

AI Verification

The finding is accurate as AES ECB mode is indeed the default mode returned by Cipher.getInstance("AES") and it's known to be weak due to its deterministic nature.

App can be installed on a vulnerable unpatched Android version 4.4-4.4.4, [minSdk=19]

HIGH

MASVS: MASVS-PLATFORM-3 CVSS: 6.8 File: AndroidManifest.xml AI CONFIRMED

Description

This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version $\Rightarrow$ 10, API 29 to receive reasonable security updates.

Business Impact

Outdated SDK targets or platform API misuse expose the app to known exploits.

AI Verification

The finding is not about a specific vulnerability in the app's code, but rather an installation issue that can be exploited by installing the app on an older Android version with known vulnerabilities. This is a valid concern as per MASVS-PLATFORM-3, which emphasizes secure use of platform APIs.

Application Data can be Backed up [android:allowBackup=true]

HIGH

MASVS: MASVS-STORAGE-1 CVSS: 6.2 File: AndroidManifest.xml AI CONFIRMED

Description

This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.

Business Impact

Sensitive user data could be extracted from the device by an attacker with physical or backup access.

AI Verification

The finding is accurate, as the `android:allowBackup` flag does indeed allow anyone to backup application data via ADB, which can be a security concern. This is a valid vulnerability in the context of mobile app security.

The App logs information. Sensitive information should never be logged.

MEDIUM

MASVS: MASVS-STORAGE-1**CVSS:** N/A**File:** sg/vantagepoint/uncrackable1/a.java **AI CONFIRMED**

Description

The App logs information. Sensitive information should never be logged.

Business Impact

Application data may be exposed through insecure local storage mechanisms.

AI Verification

Although the MASVS control is 'Secure storage of sensitive data', the finding suggests that sensitive information is being logged, which is a separate concern. Logging sensitive information can lead to unauthorized access or exposure. Even if the app doesn't store sensitive data insecurely, logging it can still be a problem.

MASVS v2 Compliance Matrix

1 controls passed | 3 controls failed | 16 controls not tested

CONTROL	DESCRIPTION	CATEGORY	STATUS
MASVS-STORAGE-1	Secure storage of sensitive data	Data Storage	FAIL
MASVS-STORAGE-2	Prevention of sensitive data leakage	Data Storage	NOT TESTED
MASVS-CRYPTO-1	Use of strong cryptographic algorithms	Cryptography	FAIL
MASVS-CRYPTO-2	Proper key management	Cryptography	NOT TESTED
MASVS-AUTH-1	Secure authentication mechanisms	Authentication & Authorization	NOT TESTED
MASVS-AUTH-2	Session management	Authentication & Authorization	NOT TESTED
MASVS-AUTH-3	Authorization controls	Authentication & Authorization	NOT TESTED
MASVS-NETWORK-1	Secure network communication (TLS)	Network Communication	NOT TESTED
MASVS-NETWORK-2	Certificate validation	Network Communication	NOT TESTED
MASVS-PLATFORM-1	Secure use of IPC mechanisms	Platform Interaction	NOT TESTED
MASVS-PLATFORM-2	Secure use of WebViews	Platform Interaction	NOT TESTED
MASVS-PLATFORM-3	Secure use of platform APIs	Platform Interaction	FAIL
MASVS-CODE-1	Input validation	Code Quality	NOT TESTED
MASVS-CODE-2	Secure coding practices	Code Quality	NOT TESTED
MASVS-CODE-3	Third-party library security	Code Quality	NOT TESTED
MASVS-CODE-4	Protection of sensitive data in memory	Code Quality	NOT TESTED

MASVS - RESILIENCE-1	Anti-debugging protection	Anti-Tampering & Reverse Engineering	PASS
MASVS - RESILIENCE-2	Root/Jailbreak detection	Anti-Tampering & Reverse Engineering	NOT TESTED
MASVS - RESILIENCE-3	Code obfuscation	Anti-Tampering & Reverse Engineering	NOT TESTED
MASVS - RESILIENCE-4	Integrity verification	Anti-Tampering & Reverse Engineering	NOT TESTED

Methodology and Tool Stack

This audit follows the OWASP Mobile Application Security Verification Standard (MASVS) v2 and the Mobile Application Security Testing Guide (MASTG).

STAGE	TOOL	DESCRIPTION
Static Analysis	MobSF	Automated decompilation and rule-based vulnerability detection
MASVS Mapping	Custom Engine	50+ rules mapping findings to MASVS v2 controls
Risk Scoring	CVSS v3.1	Standardized vulnerability scoring
AI Triage	LLM	Contextual false-positive filtering
AI Remediation	LLM + Critic	Automated fix generation with validation pass
SCA / SBOM	OSV.dev	Third-party dependency vulnerability lookup

References

- OWASP MASVS v2.0 — <https://mas.owasp.org/MASVS/>
- OWASP MASTG — <https://mas.owasp.org/MASTG/>
- CVSS v3.1 — <https://www.first.org/cvss/v3.1/specification-document>

Appendix

A. Scan Metadata

File Name	UnCrackable-Level1.apk
SHA-256	1da8bf57d266109f9a07c01bf7111a1975ce01f190b9d914bcd3ae3dbef96f21
Scan Date	May 01, 2026
Score / Grade	65.0 / 100 (C)
Findings	6 total (4 confirmed, 2 dismissed)

B. Dismissed Findings (False Positives)

#	TITLE	SEVERITY	MASVS	DISMISSAL REASON
1	This App may have root detection capabilities.	INFO	MASVS-RESILIENCE-1	The finding states that the App 'may' have root detection capabilities, but no code context is provided to support this
2	This App may request root (Super User) privileges.	INFO	MASVS-RESILIENCE-1	The finding is likely a false positive because the MASVS-RESILIENCE-1 control refers to anti-debugging protection, which

Generated by MASVS Audit Copilot — May 01, 2026
Based on OWASP MASVS v2 and MASTG Standards
This report is confidential and intended for the application owner only.